

Decoding the DNA of World-Class Universities

A comprehensive Case Study utilizing SQL-based Trend Analysis and Power BI Strategic Dashboards

1.Executive Summary

Navigating the Global Academic Arms Race

Strategic Positioning in an Era of High Saturation and Elite Moats

1.1 The Objective

This analysis evaluates over a decade of global university ranking data to identify the drivers of institutional prestige. The goal is to move beyond "rank-chasing" and define a data-driven blueprint for sustainable growth in a hyper-competitive global market.

1.2 Key Findings

- **The Saturated Middle:** Since 2014, the "middle tier" (scores of 40–50) has expanded by **900%**. Most universities are now trapped in a "competitive plateau" where incremental gains no longer result in rank increases because the entire sector is rising simultaneously.
- **The Formula for Elite Status:** Analysis of the "90+ Club" reveals a consistent "Gold Standard" DNA: a **Student-Staff Ratio (SSR) below 10:1** paired with **International Diversity above 30%**. Research Citations remain the primary "Great Divider," with top-25 schools averaging a **93.2** in research impact.
- **National Ecosystems & Efficiency:** While the U.S. and China dominate in volume, **Singapore and Hong Kong** represent the world's most "efficient" systems, producing the highest average institutional scores through hyper-focused resource allocation.

1.3 Strategic Pillars for Institutional Growth

Pillar	Strategic Reality	Recommendation
Faculty Density	High SSR (>15:1) is the #1 anchor dragging down mid-tier scores.	Aim for an SSR below 12:1 to break into the global top 100.
Internationalization	Acts as a "performance buffer" for institutions with limited research budgets.	Prioritize a 30% international student/faculty threshold to enhance global brand.
Research Output	Citations are the only metric that grants access to the Top 25.	Shift from "volume of papers" to "high-impact citations" in specialized niches.

Market	Methodology shifts (like 2011/2016)	Benchmark against percentile standing
Resilience	create artificial volatility.	rather than raw scores to gauge true progress.

1.4 The "Rising Star" Roadmap

The data proves that rapid ascent is possible. "Leapfrog" institutions like the **University of Queensland** have achieved 30+ point gains by refusing to compete on every metric. Instead, they focused on **triangulated excellence**: matching the research output of the West with the operational efficiency of the East.

1.5 Conclusion

To survive the "Crowded Middle," institutions must stop viewing rankings as a scoreboard and start viewing them as a resource-allocation map. The path to the top is not through general improvement, but through aggressive differentiation in **Faculty Density** and **Global Research Citations**.

Section 2: Strategic Recommendations

Based on the synthesis of 12 EDA insights and dual-dashboard performance metrics, the following three-step action plan is proposed for institutions seeking to break the "Competitive Plateau."

Phase 1: Break the "50-Point Barrier" through Differentiation

The Problem: EDA 2 and 3 show a massive saturation point in the 40–50 score range. Incremental demographic changes (like slightly increasing student numbers) are no longer enough to move the rank.

- **The Action:** Shift focus from **quantity to quality**. Institutions must identify a "niche of excellence." Instead of trying to improve every metric by 1%, the data suggests focusing on a 20% improvement in a single high-impact pillar (e.g., Citations or Research).
- **The Target:** Move from the crowded middle-tier into the 60+ tier, where the competitive density drops by over 75%, providing a massive visibility dividend.

Phase 2: Optimize the "Efficiency Ratio" (The Staffing-Diversity Pivot)

The Problem: EDA 9 and 10 confirm that the "Gold Standard" for elite status is a Student-Staff Ratio (SSR) below 10:1. However, many mid-tier institutions lack the immediate budget for massive hiring.

- **The Action:** Use **Internationalization as a Buffer**. The data shows that high international diversity can partially compensate for a higher SSR.
- **The Target:** Aggressively recruit international students and faculty to reach a **30% diversity threshold**. This improves the "International Outlook" score and generates the revenue needed to eventually lower the SSR to the competitive benchmark of **12:1**.

Phase 3: Transition to "Impact-First" Research

The Problem: EDA 11 and Visual 1 reveal that Citations are the "Great Divider." You cannot enter the Top 25 without an elite citation score (avg. 93.2), yet this is the hardest metric to move.

- **The Action:** Implement **Percentile Benchmarking** (as enabled by Dashboard 2). Instead of chasing raw scores, departments should be incentivized based on their standing relative to global peers.
- **The Target:** Move research strategy away from "volume of publication" toward "citation impact." High-impact research in specialized fields creates the "resource moat" identified in EDA 3, protecting the university's elite status from new market entrants.

Section 3. Exploratory Data Analysis (EDA) & Visualization Insights

Part A: The Global Landscape & Demographic Baselines

EDA 1: Institutional Shape & Descriptive Statistics

- **The Action:** Performed a statistical audit of core numerical columns (\$num_students\$, \$pct_international\$, and \$pct_female\$) to establish global baselines.
- **The Finding:** The average university size is ~24,010 students, but a massive Standard Deviation of ~13,700 reveals extreme heterogeneity—ranging from niche colleges to "mega-universities" exceeding 127,000 students. Gender parity is a global norm (~49% female), while international student ratios average ~19.7%.
- **The Strategic Insight:** Mid-tier universities must avoid benchmarking against the "Global Average." Instead, they should peer-benchmark against institutions within their same size-quartile to ensure staff-to-student ratios and resource allocations are realistic.

EDA 2: Competitive Density & Score Distribution

- **The Action:** Segmented universities into decile-based "Score Buckets" to map the competitive landscape.
- **The Finding:** There is a "Saturation Point" in the 40–50 point range, housing over 2,150 entries. In contrast, "Elite Scarcity" is real: only ~1.3% of the dataset breaks the 80-point "High Excellence" barrier.
- **The Strategic Insight:** Moving from a score of 45 to 55 provides a massive "visibility dividend," as the number of competitors drops by 75%. This jump requires shifting from incremental demographic changes to fundamental improvements in research quality.

EDA 3: Annual Score Trends and Global Competition

- **The Action:** Analyzed longitudinal trends to identify historical shifts and the impact of ranking expansions.
- **The Finding:** A massive "Market Expansion" occurred between 2014 and 2015, where middle-tier schools grew from ~100 to 900+. Additionally, a "2011 Pivot" saw average scores jump into the 50s, likely due to ranking methodology changes.
- **The Strategic Insight:** The "Elite" tier (90+ points) is protected by a resource moat. Since the top tier remains stagnant at 1–2 schools per year despite the influx of new competitors, "Elite"

status is likely a factor of massive funding and legacy research output that cannot be easily disrupted.

Part B: Regional Power & National Efficiency

EDA 4: Regional Distribution of Institutions

- **The Action:** Identified geographical concentrations of academic power.
- **The Finding:** The U.S. dominates with 273 institutions (nearly triple second-place China at 96). However, beyond the top 15 countries, the count drops sharply to only 1 or 2 schools per nation.
- **The Strategic Insight:** High-volume nations (US/China) benefit from "ecosystem advantages."¹
For schools in low-representation nations, the most viable strategy is "Regional Championship"—securing the #1 domestic spot to capture local government funding.

EDA 5: Ranking System Coverage by Region

- **The Action:** Evaluated the exclusivity of different ranking systems (CWUR vs. Shanghai vs. Times).
- **The Finding:** CWUR is the most inclusive (tracking 231 US schools), whereas the Shanghai Ranking is the "Elite Filter," focusing on a narrow slice of research-intensive giants.
- **The Strategic Insight:** Choose your battleground. Mid-tier schools should lead their marketing with CWUR rankings to gain initial global visibility before attempting the research-heavy requirements of the Shanghai index.

EDA 6: Global Average Score Trends

- **The Action:** Calculated the moving baseline of "Global Excellence."
- **The Finding:** Global averages were stable (~37) until 2011 and 2016, where dramatic spikes to 54 and 67 points occurred.
- **The Strategic Insight:** Ranking systems frequently "move the goalposts." A university's growth must exceed the global trend; if your score is flat while the average rises, you are effectively regressing.

Part C: The Anatomy of Success (Rising Stars & Elite Pillars)

EDA 8: Institutional Improvement and "Rising Stars"

- **The Action:** Calculated the "Score Delta" between the earliest and latest recorded years.
- **The Finding:** The University of Queensland and the University of Hong Kong are the "Leapfroggers," with gains of 30+ points. Most other improvers average between 12 and 22 points.
- **The Strategic Insight:** Rate of progress is a more critical KPI than current rank for mid-tier schools. Analyzing the "Queensland Model" provides a blueprint for rapid development through high-impact research investment.

EDA 9: National Efficiency and Resource Correlation

- **The Action:** Cross-analyzed average scores against Student-Staff Ratios (SSR).
- **The Finding:** Singapore is the "Efficiency Leader" (60.94 average score with a 15.8 SSR). Japan sets the gold standard for faculty density (6:1 SSR), while the UK maintains high scores despite higher student loads.

- **The Strategic Insight:** To be competitive (55+ points), aim for an SSR below 15:1. High-prestige nations are those that satisfy all three ranking systems simultaneously ("Triangulated Excellence").

EDA 10: The Intersection of Staffing and Diversity

- **The Action:** Analyzed how SSR and International % interact to drive scores.
- **The Finding:** The "Gold Standard Peak" (93.8 points) is only achieved by schools with Low SSR (<10:1) AND High International Diversity (>30%).
- **The Strategic Insight:** Internationalization can act as a "buffer." If you cannot afford to hire more staff to lower your SSR, increasing international recruitment can partially compensate and keep your ranking competitive.

EDA 11: The Anatomy of an Elite University

- **The Action:** Isolated the Top 25 universities to identify their "Core Pillars."
- **The Finding:** Citations are the "Great Divider," with an average of 93.2. Surprisingly, International Outlook is the lowest-scoring pillar (61.6) for even the top schools.
- **The Strategic Insight:** Citations are non-negotiable for the Top 25. However, because the elites are "weaker" in the International category, mid-tier schools can use high global diversity as a unique competitive edge.

EDA 12: Percentile Benchmarking of Global Leaders

- **The Action:** Translated raw scores into PERCENT_RANK() to show relative standing.
- **The Finding:** Caltech and MIT dominate the 99th percentile across all metrics, but Stanford (2011) showed a lower international percentile. The Oxford/Cambridge model is more "globally balanced."
- **The Strategic Insight:** Percentile analysis allows a board to say, "We are Top 5% in Research but only Top 40% in Internationalization," making budget priorities immediately obvious.

Part D: Power BI Dashboard Visualizations

Visual 1: Competitive Benchmarking (Top 10)

- **Analysis:** A Clustered Column Chart comparing the Top 10 across all five dimensions.
- **Key Insight:** European schools (ETH Zurich) leverage high International scores to stay in the Top 10, whereas US schools (Caltech) rely almost exclusively on perfect Research/Citation scores.

Visual 2: Country-wise Performance (Treemap)

- **Analysis:** A Treemap visualizing average total score by country size and color intensity.
- **Key Insight:** Small tiles with high intensity (Singapore/Hong Kong) prove that quality-focused national strategies outperform massive, decentralized systems.

*While the previous analysis focused on global trends, the second dashboard shifts to **Institutional Strategy**, allowing individual universities to benchmark their "Vital Signs" against the market.*

Visual 3: The Scatter Plot: Strategic Positioning

- **The Insight:** By plotting **Teaching vs. Research scores**, this visual identifies if a university is a "balanced" institution or a "specialized" one.

- **The Strategic Value:** It helps an institution identify their "cluster." Are they in the elite top-right quadrant, or are they lost in the high-density "Middle-Tier" cluster? This determines if the university needs to fix a specific weakness or maintain broad excellence.

Visual 4: KPI Cards: Operational Benchmarking

- **The Insight:** These cards provide an immediate comparison between an institution's metrics (like Student-Staff Ratio) and the **Global Average**.
- **The Strategic Value:** These are the university's "**Vital Signs**." They reveal at a glance if the school has the raw ingredients—sufficient staffing and diversity—necessary to fuel a climb in the rankings.

Visual 5: The Percentile Column Chart: Relative Strength

- **The Insight:** This chart translates abstract scores (e.g., a "70" in Citations) into a **Competitive Percentile**.
- **The Strategic Value:** A raw score of 70 is meaningless without context. If that 70 only places the school in the 50th percentile, the university is merely "average." This visual forces stakeholders to look at **relative standing** rather than just raw numbers.

Visual 6: The Dual-Line Chart: Performance vs. Market Trend

- **The Insight:** Tracks institutional progress against the **moving global baseline** over time.
- **The Strategic Value:** It answers the critical question: *"Are we actually improving, or is the whole world just getting easier?"* If a university's score remains flat while the global average rises, the school is effectively falling behind in real-time.

Visual 7 : Search & Year Slicers: Temporal & Granular Control

- **The Insight:** Allows for longitudinal analysis and specific institutional lookups over a decade.
- **The Strategic Value:** This turns a static report into a **Management Tool**. It allows boards to see if an institution's excellence is a sustainable, decade-long trend or a one-time methodology spike.

Part D.2 : Synthesis — "The So What?"

The project culminates in two distinct strategic views:

1. **Dashboard 1 (Macro View):** Designed for **Government and Industry** levels to understand the global academic arms race, regional power shifts, and national efficiency.
2. **Dashboard 2 (Micro View):** Designed for the **University Board** level. It moves from "What happened?" to "What should we do?" by identifying exactly where to allocate next year's budget—whether that is hiring more staff (lowering SSR) or investing in citation-heavy research.

Section 4: Technical Documentation

This section provides the underlying logic and data transformation steps that powered the EDA and Dashboards. By utilizing complex joins, window functions, and Common Table Expressions (CTEs), we translated raw university data into actionable strategic insights.

1. Data Dictionary

This section serves as the technical blueprint for the relational schema, defining every column used in the SQL analysis and Power BI model.

1. Table: `university`

The master dimension table containing institutional identities.

- **id (PK):** `INT` – Unique identifier for each university.
- **country_id (FK):** `INT` – Reference to the `country.id`.
- **university_name:** `VARCHAR` – The official, standardized name of the institution (UTF-8 encoded).

2. Table: `country`

Geographical mapping for regional analysis.

- **id (PK):** `INT` – Unique identifier for each country.
- **country_name:** `VARCHAR` – The full name of the country.

3. Table: `university_year`

The institutional demographic fact table.

- **university_id (FK):** `INT` – Reference to `university.id`.
- **year (PK):** `INT` – The specific academic year for the record.
- **num_students:** `INT` – Total student enrollment.
- **student_staff_ratio:** `DECIMAL` – The number of students per full-time academic staff member.
- **pct_international_students:** `FLOAT` – Percentage of the student body with international status.
- **pct_female_students:** `FLOAT` – Percentage of the student body identifying as female.

4. Table: `ranking_system`

The source of the ranking data.

- **id (PK):** `INT` – Unique identifier for the ranking body.
- **system_name:** `VARCHAR` – Name of the system (e.g., *Times Higher Education*, *Shanghai Ranking*, *CWUR*).

5. Table: `ranking_criteria`

Defines the specific pillars of academic excellence.

- **id (PK):** INT – Unique identifier for the specific metric.
- **ranking_system_id (FK):** INT – Reference to ranking_system.id.
- **criteria_name:** VARCHAR – The metric name (e.g., *Citations, Teaching, Research, Total Score*).

6. Table: university_ranking_year

The core fact table containing performance scores.

- **university_id (FK):** INT – Reference to university.id.
- **ranking_criteria_id (FK):** INT – Reference to ranking_criteria.id.
- **year (PK):** INT – The year the ranking was issued.
- **score:** FLOAT – The numerical score achieved (typically on a scale of 0–100).

2. Data Cleaning & Preprocessing

Objective: To transform raw, fragmented university records into a "Single Source of Truth" by addressing inconsistencies, missing values, and referential integrity gaps.

A. Structural Integrity & Uniqueness

- **The Action:** Audited Primary Keys across *university_year* and *university_ranking_year* to identify duplicates.
- **The Result:** Verified unique institutional records per year, preventing the inflation of average scores in the final dashboard.
- **Referential Integrity:** Performed **Left Joins** to detect "orphaned" records (rankings without matching master university entries). This ensured the Power BI relational model remained synchronized.

B. Handling Missing Data (Imputation vs. Deletion)

- **Selective Imputation:** For the *pct_female_students* column, **Mean Imputation** filled null values. Since gender parity is a critical demographic metric, using the global mean ($\bar{x}$) preserved dataset volume without introducing significant bias.
- **Strategic Deletion:** In the *university_ranking_year* table, records with missing *score* values were purged. Given the low frequency of missing scores, deletion maintained the precision of "Top University" rankings more effectively than synthetic estimation.

C. Domain Validation & Outlier Detection

- **Logical Constraints:** Implemented range checks ensuring percentages (International/Female) were strictly between $[0, 100]$ and population counts were non-negative ($n > 0$).
- **Statistical Spread:** Calculated Min, Max, and Standard Deviation (σ) to identify extreme outliers in the **Student-Staff Ratio (SSR)** that could skew "Gold Standard" benchmarks.

D. Text Standardization & Character Encoding

- **The Problem:** The `university_name` field contained "mojibake" (corrupted characters) such as 'Pontifical Catholic University of ParanÃi'.
- **The Fix:** Used SQL regex to detect non-ASCII characters and performed a Binary Conversion (**latin1 to utf8mb4**) to restore the linguistic integrity of global institution names.

3. Engineering Best Practices

To ensure the analysis met industry standards for data engineering, the following protocols were enforced:

- **Deduplication:** String normalization was applied to ensure entities like "Harvard" and "Harvard University" were consolidated.
- **Reproducibility:** All cleaning steps were executed via **SQL scripts** rather than manual edits, creating an immutable audit trail.
- **Environment Security:** Mass updates were handled using `SET sql_safe_updates = 0;` within a controlled session to ensure database safety during large-scale transformations.
- **Type Consistency:** Percentage columns were cast as **floats/decimals** rather than strings, allowing for complex mathematical aggregation in DAX and SQL.

4. Analytical Logic (SQL & DAX)

1. Data Foundation & Distribution

To establish the "baseline" of the dataset, SQL was used to audit the shape of the numerical data and categorize institutions into performance tiers.

- **Metric Audit:** We calculated μ (Mean) and σ (Standard Deviation) to identify outliers and the high degree of heterogeneity between "Mega-Universities" and "Niche Colleges."
- **Score Tiering:** Using $\text{FLOOR}(\text{score}/10) * 10$, we generated the distribution buckets that revealed the "Saturation Point" in the 40–50 range.

SQL

```
-- Logic for Bucket-wise Score Distribution
SELECT year, FLOOR(score/10) * 10 AS score_bucket, count(*)
FROM university_ranking_year ury
JOIN ranking_criteria rc ON ury.ranking_criteria_id = rc.id
WHERE rc.criteria_name LIKE 'Total%'
GROUP BY 1,2 ORDER BY 1,2;
```

2. Longitudinal Performance & "Rising Star" Analysis

Identifying "Rising Stars" required a complex CTE to isolate the first and last recorded scores for every university to calculate the total improvement delta.

- **Window Functions:** Used FIRST_VALUE and LAST_VALUE over a partition of university_id to ignore annual methodology noise and focus on long-term trajectory.
- **SSR Volatility:** The LAG() function was utilized to track how faculty density changed year-over-year, identifying if rank improvements were correlated with staffing investments.

SQL

```
-- Excerpt: Logic for Largest Improvement (Rising Stars)
WITH yearly_avg AS (
  SELECT university_id, year, AVG(score) AS avg_score
  FROM university_ranking_year ury
  JOIN ranking_criteria rc ON ury.ranking_criteria_id = rc.id
  WHERE rc.criteria_name LIKE 'Total%'
  GROUP BY university_id, year
)
SELECT u.university_name, ly.avg_score - fy.avg_score AS score_improvement
FROM yearly_avg fy
JOIN yearly_avg ly ON fy.university_id = ly.university_id
JOIN university u ON u.id = fy.university_id
WHERE fy.year = (SELECT MIN(year) FROM yearly_avg y2 WHERE y2.university_id = fy.university_id)
  AND ly.year = (SELECT MAX(year) FROM yearly_avg y3 WHERE y3.university_id = fy.university_id)
ORDER BY score_improvement DESC LIMIT 10;
```

3. Dimensional Correlation: Staffing vs. Diversity

A key strategic discovery was the interaction between Student-Staff Ratios (SSR) and Internationalization. We used a CASE statement matrix to group universities into "Staffing Groups" and "Diversity Groups."

- **Strategic Findings:** This query proved that schools with **High Internationalization (>30%)** outperformed peers even when their staffing levels were sub-optimal.

SQL

```
-- SSR vs. Internationalization Matrix
SELECT
  CASE WHEN uy.student_staff_ratio < 10 THEN 'Low (<10: Best)'
  WHEN uy.student_staff_ratio BETWEEN 10 AND 15 THEN 'Medium (10–15)'
  ELSE 'High (>15: Worst)' END AS staff_ratio_group,
  CASE WHEN uy.pct_international_students > 30 THEN 'High intl %(>30)'
  WHEN uy.pct_international_students < 15 THEN 'Low intl %(<15)'
  ELSE 'Moderate (15-30)' END AS int_pct_grp,
  AVG(ury.score) AS avg_score
FROM university_year uy
JOIN university_ranking_year ury ON uy.university_id = ury.university_id AND uy.year = ury.year
WHERE ury.ranking_criteria_id IN (SELECT id FROM ranking_criteria WHERE criteria_name LIKE 'Total%')
```

```
GROUP BY 1,2 ORDER BY avg_score DESC;
```

4. Competitive Percentile Benchmarking

To support **Dashboard 2**, we transitioned from raw scores to PERCENT_RANK(). This provides a relative standing of an institution against the entire global population for each specific pillar (Citations, Teaching, Research).

SQL

```
-- Logic for Relative Rank (Percentile)
SELECT u.university_name, rs.year,
       PERCENT_RANK() OVER (ORDER BY rs.overall_score) AS overall_percentile,
       PERCENT_RANK() OVER (ORDER BY rs.citation_score) AS citation_percentile,
       PERCENT_RANK() OVER (ORDER BY rs.research_score) AS research_percentile
FROM ranked_score rs
JOIN university u ON rs.university_id = u.id
ORDER BY overall_percentile DESC LIMIT 50;
```

POWER BI - DAX Queries

1. Performance Normalization & Averages

Due to the methodology shifts identified in EDA 6, the following measures normalize the performance data from 2011 onwards to provide a consistent baseline for analysis.

- **Average Total Score:** Calculates a standardized mean across all "Total" ranking criteria while applying a historical filter to ensure data relevance.
- **TREATAS Logic:** Implemented for complex cross-table filtering, allowing demographic data (SSR/International %) to bridge over to ranking scores seamlessly.

Code snippet

Average Total Score =

```
CALCULATE(
    AVERAGE('universitydb university_ranking_year'[score]),
    CONTAINSSTRING('universitydb ranking_criteria'[criteria_name], "Total"),
    KEEPFILTERS('universitydb university_ranking_year'[year] >= 2011)
)
```

2. Competitive Percentile Ranking

To support the **"Relative Strength"** visual in Dashboard 2, we utilized a virtual table approach. This calculates a university's standing compared to all other institutions in the league for a specific year and ranking system.

- **Logic:** The measure creates a virtual summary of all scores (SUMMARIZECOLUMNS), removes the university slicer filter (ALL), and then counts how many institutions fall below the selected university's score to find its percentile rank.

Code snippet

Citation Percentile Rank =

```
VAR OurScore = CALCULATE(AVERAGE('universitydb university_ranking_year'[score]), 'universitydb
ranking_criteria'[id] = 4)
VAR AllScoresSystem =
    SUMMARIZECOLUMNS(
        'universitydb university_ranking_year'[university_id],
        'universitydb university_ranking_year'[year],
        ALL('universitydb university'),
        "@Score", AVERAGE('universitydb university_ranking_year'[score])
    )
VAR CountBelowOrEqual = COUNTROWS(FILTER(AllScoresSystem, [@Score] <= OurScore))
VAR TotalCount = COUNTROWS(AllScoresSystem)
RETURN DIVIDE(CountBelowOrEqual, TotalCount) * 100
```

3. High-Impact Strategic Indicators

These measures drive the "So What?" factors of the dashboard, providing the Board with clear targets and historical momentum.

- **Score Improvement:** Identifies the "Rising Star" factor by calculating the delta between an institution's very first and very last recorded "Total" score.
- **Target Top 10 Average:** Dynamically identifies the top 10 scoring institutions globally (ignoring the current university selection) to provide a "north star" benchmark for leadership.

Code snippet

Target Top 10 Average Score =

```
VAR AllUnivScores =
    CALCULATETABLE(
        ADDCOLUMNS(SUMMARIZE(ALL('universitydb university'), 'universitydb
university'[university_name]), "@Score", [Average Total Score]),
        'universitydb university_ranking_year'[year] >= 2011,
        CONTAINSSTRING('universitydb ranking_criteria'[criteria_name], "Total")
    )
VAR Top10Scores = TOPN(10, AllUnivScores, [@Score], DESC)
RETURN AVERAGEX(Top10Scores, [@Score])
```

4. Dynamic Visual Styling (Conditional Formatting)

To improve user experience, DAX was used to manage the **Scatter Plot's** visual hierarchy.

- **Scatter Color/Size:** Uses SELECTEDVALUE and MAX to highlight the user's selected university in high-contrast red (#E53935) while dimming all other data points to a subtle grey.

5. The Relationship Bridge: TREATAS Total Score

This is a high-level measure used when you need to apply filters across tables that aren't directly related in the schema, specifically linking demographic data to ranking scores while enforcing a "2011 onward" rule.

Code snippet

Average Total Score TRETAS =

```
VAR UnivYearFilter =
    SELECTCOLUMNS(
        'universitydb university_year',
        "university_id", 'universitydb university_year'[university_id],
        "year",          'universitydb university_year'[year]
    )
RETURN
CALCULATE(
    AVERAGE('universitydb university_ranking_year'[score]),
    -- The "TREATAS" Magic: maps the virtual table of IDs and Years
    -- directly onto the Ranking table to enforce strict alignment.
    TREATAS(
        UnivYearFilter,
        'universitydb university_ranking_year'[university_id],
        'universitydb university_ranking_year'[year]
    ),
    FILTER(
        'universitydb ranking_criteria',
        CONTAINSSTRING('universitydb ranking_criteria'[criteria_name], "Total")
    ),
    'universitydb university_ranking_year'[year] >= 2011
)
```

2. The Benchmark: Dynamic "Top 10" Target

To provide a goal for university boards, this measure calculates the average score of the global top 10 institutions. It is complex because it must ignore the university the user has selected (using ALL) to maintain an objective target.

Code snippet

```

Target Top 10 Average Score =
VAR CurrentCriteriaName = "Total"
-- Create a virtual table of ALL universities globally
VAR AllUnivScores =
    CALCULATETABLE(
        ADDCOLUMNS(
            SUMMARIZE(
                ALL('universitydb university'), -- Force ignore the slicer
                'universitydb university'[university_name]
            ),
            "@Score", [Average Total Score]
        ),
        'universitydb university_ranking_year'[year] >= 2011,
        CONTAINSSTRING('universitydb ranking_criteria'[criteria_name], CurrentCriteriaName)
    )
-- Select only the top 10 from that virtual list
VAR Top10Scores =
    TOPN(10, AllUnivScores, [@Score], DESC)
-- Average those 10 values
RETURN
    AVERAGEX(Top10Scores, [@Score])

```

3. The Relative Standing: Citation Percentile Rank

This measure converts a raw score into a percentile. It is a "heavy" measure because it creates a virtual league table in memory for every year and university to find the selected institution's rank.

Code snippet

```

Citation Percentile Rank =
VAR CriteriaID = 4 -- ID for 'Citations'
VAR OurScore =
    CALCULATE(
        AVERAGE('universitydb university_ranking_year'[score]),
        'universitydb ranking_criteria'[id] = CriteriaID
    )
-- Build the Global League Table virtually
VAR AllScoresSystem =
    SUMMARIZECOLUMNS(
        'universitydb university_ranking_year'[university_id],
        'universitydb university_ranking_year'[year],
        ALL('universitydb university'), -- Ignore individual university filter
    )

```

```

        "@Score", AVERAGE('universitydb university_ranking_year'[score])
    )
-- Count how many schools we have out-performed
VAR CountBelowOrEqual =
    COUNTROWS(FILTER(AllScoresSystem, [@Score] <= OurScore))
VAR TotalCount = COUNTROWS(AllScoresSystem)
RETURN
    DIVIDE(CountBelowOrEqual, TotalCount) * 100

```

4. The Longitudinal Insight: Score Improvement (Delta)

This measure tracks a university's total growth. It dynamically finds the earliest year they entered the rankings and their most recent year, then subtracts the scores.

Code snippet

```

Score Improvement =
VAR BaseTable =
    FILTER(
        ADDCOLUMNS(
            'universitydb university_ranking_year',
            "CriteriaName", RELATED('universitydb ranking_criteria'[criteria_name])
        ),
        CONTAINSSTRING([CriteriaName], "Total")
    )
-- Dynamically find the start and end points for THIS specific university
VAR FirstYear = CALCULATE(MIN('universitydb university_ranking_year'[year]), BaseTable)
VAR LastYear = CALCULATE(MAX('universitydb university_ranking_year'[year]), BaseTable)
VAR FirstScore = CALCULATE(AVERAGE('universitydb university_ranking_year'[score]), BaseTable,
    'universitydb university_ranking_year'[year] = FirstYear)
VAR LastScore = CALCULATE(AVERAGE('universitydb university_ranking_year'[score]), BaseTable,
    'universitydb university_ranking_year'[year] = LastYear)
RETURN LastScore - FirstScore

```

5. The User Experience: Scatter Highlight Logic

These two measures work together to make the scatter plot interactive. When a user selects a university in a slicer, that one dot turns red and grows in size, while others fade.

Code snippet

```

-- Logic for Color Highlighting
Scatter Color =
VAR SelectedUniv = SELECTEDVALUE('University Selector'[university_name])
VAR CurrentUniv = MAX('universitydb university'[university_name])

```

```

RETURN
IF(
    NOT ISBLANK(SelectedUniv) && CurrentUniv = SelectedUniv,
    "#E53935", -- Strategic Red for selected
    "#BDBDBD" -- Neutral Grey for others
)
-- Logic for Dot Size
Scatter Size =
VAR SelectedUniv = SELECTEDVALUE('University Selector'[university_name])
RETURN
IF(
    MAX('universitydb university'[university_name]) = SelectedUniv,
    3, -- Emphasized Size
    0.1 -- Background Size
)

```

Final Project Conclusion

The global academic landscape is no longer a static hierarchy; it is a dynamic, high-stakes competition. This case study proves that while the "Elite Moat" is difficult to penetrate, it is not invincible. By leveraging **staffing efficiency, international diversity, and high-impact citations**, any institution can use this data-driven roadmap to move from a localized player to a global "Rising Star."